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Lithium ion secondary battery and method for manufacturing same

Abstract

A lithium ion secondary battery with increased durability and capacity includes a positive electrode, a negative electrode, and a separator. The positive electrode includes a current collector foil and an electrode mixture layer disposed on a surface of the current collector foil. The positive electrode mixture layer includes a superficial layer portion and a deep layer portion. The superficial layer portion opposes the negative electrode via the separator. The deep layer portion is disposed between the superficial layer portion and the current collector foil. The superficial layer portion contains positive electrode active material particles having an average particle diameter larger than an average particle diameter of positive electrode active material particles contained in the deep layer portion. A space ratio between the positive electrode active material particles in the superficial layer portion is lower than a space ratio between the positive electrode active material particles in the deep layer portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/292,293, filed May 7, 2021, which is a national phase application of International Application No. PCT/JP2019/042865, filed Oct. 31, 2019, which claims the benefit of priority from Japanese Application No. 2018-212635, filed Nov. 13, 2018, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD
(1) The present disclosure relates to a lithium ion secondary battery and a method for manufacturing the same.
BACKGROUND ART
(2) Conventionally, there has been known an invention relating to a positive electrode plate and a lithium ion secondary battery that includes the positive electrode plate (see Patent Literature 1 below). The invention disclosed in Patent Literature 1 has an object to provide a positive electrode plate that can ensure battery capacity while decreasing a lithium ion diffusion resistance in the use in a lithium ion secondary battery (see paragraph 0005 and the like in the literature). To achieve this object, Patent Literature 1 discloses an invention relating to a positive electrode plate that has the following configuration.
(3) The positive electrode plate includes a metal foil that has a first principal surface and a second principal surface and a laminated active material layer formed on at least any one of the first principal surface or the second principal surface, and a plurality of positive electrode active material layers containing positive electrode active material particles formed of a lithium compound are laminated in the laminated active material layer. The laminated active material layer has a uniform content rate of the positive electrode active material particles viewed in the lamination direction, and the smaller average particle diameter of the contained positive electrode active material particles the positive electrode active material layer has, the upper layer the positive electrode active material layer is disposed in (see claim **1** and the like in the literature).
(4) With this configuration, the lithium ion diffusion resistance can be decreased in the upper layer of the positive electrode plate, and decrease in battery capacity can be avoided while reducing the lithium ion diffusion resistance to decrease an internal resistance of the battery, thereby allowing the ensured capacity (see paragraph 0007 and the like in the literature). When the positive electrode plate is used in the battery, increase in battery internal resistance value over time can be suppressed

even when charge and discharge is repeated (see paragraph 0008 and the like in the literature).
CITATION LIST
Patent Literature
(5) Patent Literature 1: JP 2009-026599 A
SUMMARY OF INVENTION
Technical Problem
(6) In the above-described conventional positive electrode plate, as described above, the positive electrode active material layer having the smaller average particle diameter of the contained positive electrode active material particles is disposed in the upper layer. In addition, in a lithium ion secondary battery using the positive electrode plate, the positive electrode active material layer that is disposed in the upper layer of the positive electrode plate and has the small average particle diameter of the contained positive electrode active material particles is opposed to a negative electrode plate via a separator (see paragraph 0027, FIG. 3, and the like in the literature). The conventional lithium ion secondary battery having such a configuration has the following problems.
(7) The positive electrode active material layer in the upper layer of the positive electrode plate opposing the negative electrode plate via the separator increases in intercalation and deintercalation of lithium ions due to charge/discharge reaction of the battery compared with the positive electrode active material layer in the lower layer. However, for the positive electrode active material particles, cracking of the positive electrode active material particles easily occurs due to the intercalation and the deintercalation of the lithium ions as the average particle diameter relatively decreases. When the cracking of the positive electrode active material particles occurs, the internal resistance value of the battery possibly increases, or the battery capacity possibly decreases.
(8) The present disclosure provides a lithium ion secondary battery that is excellent in durability and increased in capacity compared with the conventional one, and a method for manufacturing the same.
Solution to Problem
(9) An aspect of the disclosure is a lithium ion secondary battery that includes a positive electrode, a negative electrode, and a separator. The positive electrode includes a positive electrode current collector foil and a positive electrode mixture layer disposed on a surface of the positive electrode current collector foil. The positive electrode mixture layer includes a superficial layer portion and a deep layer portion, the superficial layer portion opposes the negative electrode via the separator, and the deep layer portion is disposed between the superficial layer portion and the positive electrode current collector foil. The superficial layer portion contains positive electrode active material particles having an average particle diameter larger than an average particle diameter of positive electrode active material particles contained in the deep layer portion. A space ratio between the positive electrode active material particles in the superficial layer portion is lower than a space ratio between the positive electrode active material particles in the deep layer portion.
Advantageous Effects of Invention
(10) The one aspect of the present disclosure can provide the lithium ion secondary battery that is excellent in durability and increased in capacity compared with the conventional one.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a lithium ion secondary battery according to one embodiment of this disclosure.
(2) FIG. **2** is an exploded perspective view of the lithium ion secondary battery illustrated in FIG. **1**.

(3) FIG. **3** is an exploded perspective view of a wound body of the lithium ion secondary battery illustrated in FIG. **2**.
(4) FIG. **4** is a schematic cross-sectional view of a positive electrode constituting the wound body illustrated in FIG. **3**.
(5) FIG. **5** is a schematic enlarged cross-sectional view of the positive electrode illustrated in FIG. **4**.
(6) FIG. **6** is a flowchart of a method for manufacturing the lithium ion secondary battery according to the one embodiment of this disclosure.
(7) FIG. **7** is a schematic drawing of a manufacturing apparatus used in an applying step and a drying step illustrated in FIG. **6**.

DESCRIPTION OF EMBODIMENTS

(8) The following describes one embodiment of a lithium ion secondary battery and a method for manufacturing the same according to the present disclosure with reference to the drawings.
(9) FIG. **1** is a perspective view of a secondary battery **100** according to the one embodiment of the present disclosure. FIG. **2** is an exploded perspective view of the secondary battery **100** illustrated in FIG. **1**. FIG. **3** is an exploded perspective view of a wound body **30** of the secondary battery **100** illustrated in FIG. **2**. FIG. **4** is a schematic cross-sectional view of a positive electrode **31** constituting the wound body **30** illustrated in FIG. **3**. FIG. **5** is a schematic enlarged cross-sectional view of the positive electrode **31** illustrated in FIG. **4**.
(10) The secondary battery **100** of this embodiment is a square secondary battery used for an electric storage device of, for example, an electric vehicle (EV) and a hybrid electric vehicle (HEV). More specifically, the secondary battery **100** is, for example, a square lithium ion secondary battery. For such a secondary battery **100**, an increased capacity and an improved durability against high current input/output have been desired. While the details will be described later, the secondary battery **100** of this embodiment has the following configuration.
(11) The secondary battery **100** is a lithium ion secondary battery, and includes the positive electrode **31**, a negative electrode **32**, and separators **33**, **34**. The positive electrode **31** includes a positive electrode current collector foil **31***a* and positive electrode mixture layers **31***b* disposed on surfaces of the positive electrode current collector foil **31***a*. The positive electrode mixture layers **31***b* include superficial layer portions **31***s* opposing the negative electrode **32** via the separators **33**, **34**, and deep layer portions **31***d* disposed between the superficial layer portions **31***s* and the positive electrode current collector foil **31***a*. The superficial layer portion **31***s* contains positive electrode active material particles Ps having an average particle diameter greater than an average particle diameter of positive electrode active material particles Pd contained in the deep layer portion **31***d*. A space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* is lower than a space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*.
(12) The following describes the configurations of the respective portions of the secondary battery **100** of this embodiment in detail. In the respective drawings, in some cases, the configurations of the respective portions of the secondary battery **100** are described using an XYZ orthogonal coordinate system including X-axis parallel to the width direction of the flat square secondary battery **100**, Y-axis parallel to the thickness direction, and Z-axis parallel to the height direction. In the following description, directions of up-down and left-right are directions convenient for describing the configurations of the respective portions of the secondary battery **100** based on the drawings. They are not limited to the vertical direction or the horizontal direction.
(13) The secondary battery **100** includes, for example, a battery container **10**, external terminals **20**, the wound body **30**, current collector plates **40**, and an insulating sheet **50**. The battery container **10**, for example, is a metallic container having a flat and rectangular box shape. The battery container **10** includes a pair of wide side surfaces **10***w* along the width direction (X-direction), a pair of narrow side surfaces **10***n* along the thickness direction (Y-direction), and elongated

rectangular top surface **10***t* and bottom surface **10***b*. Among these wide side surfaces **10***w*, narrow side surfaces **10***n*, top surface **10***t*, and bottom surface **10***b*, the wide side surface **10***w* has the largest area.
(14) The battery container **10** includes, for example, a flat square battery can **11** having opened one end in the height direction (Z-direction), and a rectangular-plate-shaped battery lid **12** that closes an opening **11***a* of the battery can **11**. The wound body **30** as an electric storage element is internally inserted into the battery container **10** from the opening **11***a* of the battery can **11**. In the battery container **10**, the battery lid **12** is welded over the whole circumference of the opening **11***a* of the battery can **11** by, for example, laser beam welding, thus sealing the opening **11***a* of the battery can **11** with the battery lid **12**.
(15) On both ends in the longitudinal direction as the width direction (X-direction) of the secondary battery **100**, the battery lid **12** is provided with through holes **12***a* through which the external terminals **20** are partially inserted. The battery lid **12** includes a gas discharge valve **15** at the center portion in the longitudinal direction. The gas discharge valve **15** is a portion having a slit formed by, for example, thinning a part of the battery lid **12** by presswork, and is integrally disposed with the battery lid **12**. An increase of an internal pressure of the battery container **10** up to a predetermined pressure causes cleavage of the gas discharge valve **15** to discharge a gas inside the battery container **10**, which reduces the internal pressure of the battery container **10** to ensure the safety of the secondary battery **100**.
(16) The battery lid **12** is provided with, for example, a liquid injection hole **16** between the through hole **12***a* and the gas discharge valve **15**. The liquid injection hole **16** is provided to inject an electrolyte to an inside of the battery lid **12** and is sealed by joining a liquid injection plug **17** by, for example, laser beam welding after injecting the electrolyte. As a nonaqueous electrolyte to be injected into the battery container **10**, for example, one obtained by dissolving lithium hexafluorophosphate (LiPF.sub.6) in a mixed solution, in which ethylene carbonate and dimethyl carbonate are mixed with a volume ratio of 1:2, with a concentration of 1 mol/L can be used. Note that the nonaqueous electrolyte may contain a lithium electrolyte other than the lithium hexafluorophosphate to have a satisfactory ion conductivity, or an additive to maintain an electrode reaction activation in the electrolyte solvent.
(17) The pair of external terminals **20** are separately disposed in the longitudinal direction of an outer surface of the battery lid **12**, that is, the top surface **10***t* of the battery container **10**, pass through the battery lid **12**, and are connected to respective base portions **41** of the pair of current collector plates **40** inside the battery container **10**. The external terminals **20** include a positive electrode external terminal **20**P and a negative electrode external terminal **20**N. A material of the positive electrode external terminal **20**P is, for example, aluminum or an aluminum alloy. A material of the negative electrode external terminal **20**N is, for example, copper or a copper alloy.
(18) The external terminals **20** include, for example, joint portions **21** to be joined to busbars and connecting portions **22** to be connected to the current collector plates **40**. The joint portions **21** have rectangular block shapes in approximately rectangular parallelepiped shapes and are disposed in the outer surface of the battery lid **12**, that is, the top surface **10***t* of the battery container **10** via gaskets **13** having an electrical insulating property. The connecting portion **22** is a column-shaped or cylindrical-shaped portion extending in a direction passing through the battery lid **12** from a bottom surface of the joint portion **21** opposed to the battery lid **12**.
(19) The current collector plates **40** include the base portions **41** connected to the external terminals **20**, extending portions **42** extending in a direction intersecting the base portions **41**, and bent portions **43** disposed between joint portions **42***a* of the extending portions **42** joined to the wound body **30** and the base portions **41**. The base portions **41** are disposed along an inner surface of the battery lid **12**, and the extending portions **42** extend toward a direction perpendicular to the inner surface of the battery lid **12**. The joint portion **42***a* of the extending portion **42** is joined by, for example, ultrasonic joining to a laminated portion **35** in which a positive electrode current collector

portion **31***c* or a negative electrode current collector portion **32***c* of the wound body **30** is wound and flatly laminated.

(20) The wound body **30** includes, for example, the positive electrode **31**, the negative electrode **32**, and the first separator **33** and the second separator **34** as insulators that insulate these electrodes. The wound body **30** is a wound electrode group having a configuration where the first separator **33**, the positive electrode **31**, the second separator **34**, and the negative electrode **32** are laminated and wound. In the wound body **30**, for example, the electrode wound around the innermost periphery and the outermost periphery is the negative electrode **32**, and the first separator **33** is further wound around an outer periphery of the negative electrode **32** wound around the outermost periphery.

(21) The negative electrode **32** includes a negative electrode current collector foil **32***a*, negative electrode mixture layers **32***b* formed on both its front and back surfaces, and the negative electrode current collector portion **32***c* as a portion where the negative electrode current collector foil **32***a* is exposed from the negative electrode mixture layers **32***b*. The negative electrode current collector portion **32***c* of the negative electrode **32** is disposed on one side in the width direction (X-direction) of the long strip-shaped negative electrode **32**, that is, a winding axis direction D of the wound body **30**. As the negative electrode current collector foil **32***a*, for example, a copper foil having a thickness of approximately 6 μm to approximately 12 μm can be used, and it is preferably an electrolytic copper foil of about 8 μm.

(22) The negative electrode mixture layers **32***b* are formed by, for example, applying a slurry negative electrode mixture to both the front and back surfaces of the negative electrode current collector foil **32***a* excluding the negative electrode current collector portion **32***c*, and drying the applied negative electrode mixture and pressing them. Subsequently, the negative electrode current collector foil **32***a* on which the negative electrode mixture layers **32***b* are formed is appropriately cut, thereby allowing fabricating the negative electrode **32**. The negative electrode mixture layers **32***b* have a thickness excluding the negative electrode current collector foil **32***a* of, for example, about 70 μm.

(23) The slurry of the negative electrode mixture can be prepared, for example, as follows. A water solution of carboxymethyl cellulose (CMC) as a viscosity-adjusting agent is added to 100 parts by weight of a graphite carbon powder as a negative electrode active material, and mixed. 1 part by weight of styrene butadiene rubber (SBR) as a binder is further added to the mixture, pure water (water from which metal ions are removed by ion exchange) is further added as a dispersing solvent and mixed, and the obtained one can be used as the slurry of the negative electrode mixture.

(24) When an insulating layer is formed on the surface of the negative electrode mixture layer **32***b*, an insulation slurry in which an insulating material is dispersed can be applied over the surface of the negative electrode mixture slurry applied over the surface of the negative electrode current collector foil **32***a*. Subsequently, the slurry of the negative electrode mixture and the insulation slurry are dried and pressed, thus allowing forming the negative electrode mixture layer **32***b* that includes the insulating layer on the surface. When the insulating layer is not formed on the surface of the negative electrode mixture layer **32***b*, it is preferred to use the separators **33**, **34** that include layers containing inorganic materials, such as a ceramic layer, on the surface.

(25) Note that the negative electrode active material included in the negative electrode mixture layer **32***b* is not limited to the above-described graphitic carbon. For example, as the negative electrode active material, amorphous carbon, natural graphite capable of insertion and desorption of lithium ions, various kinds of artificial graphite materials, a graphitic carbon material on which surface modification or surface reforming is performed, a carbonaceous material, such as coke, a compound of Si, Sn, and the like (for example, SiO and TiSi.sub.2), or a composite material of these substances may be used. A particle shape of the negative electrode active material is not especially limited and may be a scaly shape, a spherical shape, a fiber shape, a lump shape, and the like. Among them, a negative electrode active material containing particles formed in spherical shapes as a main component is preferably used.

(26) The positive electrode **31** includes a positive electrode current collector foil **31***a* as a positive electrode current collector, positive electrode mixture layers **31***b* formed on both its front and back surfaces, and the positive electrode current collector portion **31***c* as a portion where the positive electrode current collector foil **31***a* is exposed from the positive electrode mixture layers **31***b*. The positive electrode current collector portion **31***c* of the positive electrode **31** is disposed on one side on a side opposite to the negative electrode current collector portion **32***c* of the negative electrode **32** in the width direction (X-direction) of the long strip-shaped positive electrode **31**, that is, the winding axis direction D of the wound body **30**. As the positive electrode current collector foil **31***a*, for example, an aluminum foil having a thickness of approximately 10 μm to approximately 20 μm can be used, and it is preferably an aluminum foil having a thickness of about 15 μm. A foil material of the positive electrode current collector foil **31***a* is more preferably a foil material having a tensile strength of 250 N/mm.sup.2 or more.
(27) As described above, the positive electrode mixture layers **31***b* include the superficial layer portions **31***s* opposing the negative electrode mixture layers **32***b* of the negative electrode **32** via the separators **33**, **34**, and the deep layer portions **31***d* disposed between the superficial layer portions **31***s* and the positive electrode current collector foil **31***a*. That is, the deep layer portion **31***d* is disposed on the surface of the positive electrode current collector foil **31***a*, and the superficial layer portion **31***s* is disposed to cover the surface of the deep layer portion **31***d*. In other words, the deep layer portion **31***d* is disposed on the positive electrode current collector foil **31***a*, and the superficial layer portion **31***s* is disposed on the deep layer portion **31***d*.
(28) While the details will be described later, the positive electrode mixture layers **31***b* are formed by, for example, laminating two types of slurry positive electrode mixtures containing two types of different positive electrode active material particles Ps, Pd, applying them to the surfaces of the positive electrode current collector foil **31***a* excluding the positive electrode current collector portion **31***c*, and drying the applied two types of positive electrode mixtures and pressing them. Subsequently, the positive electrode current collector foil **31***a* on which the positive electrode mixture layers **31***b* are formed is appropriately cut, thereby allowing fabricating the positive electrode **31**. The positive electrode **31** may include an insulating layer on the surface similarly to the negative electrode **32**.
(29) The positive electrode mixture layers **31***b* have a thickness excluding the positive electrode current collector foil **31***a* of, for example, about 70 μm. As the slurry of the positive electrode mixture, for example, one obtained as follows can be used. 7 parts by weight of scaly graphite, acetylene black, or both of them as a conductive material and 3 parts by weight of PVDF as a binder are added to 100 parts by weight of layered nickel cobalt lithium manganate (chemical formula $Li(Ni_xCo_yMn_{1-x-y})O_2$) as the positive electrode active material, and N-methyl pyrrolidone (NMP) is further added as a dispersing solvent, thus mixing them.
(30) In the chemical formula: $Li(Ni_xCo_yMn_{1-x-y})O_2$ of the layered nickel cobalt lithium manganate as the positive electrode active material, x and y are preferably in a range of $0.30<x\leq 0.85$ and $0.18<y<0.40$. For example, as the positive electrode active material, the layered nickel cobalt lithium manganate indicated by the chemical formula: $Li(Ni_{0.6}Co_{0.2}Mn_{0.2})O_2$ can be used. The cobalt content of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is higher than, for example, the cobalt content of the positive electrode active material particles Pd contained in the deep layer portion **31***d*.
(31) As described above, the average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is greater than the average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d*. More specifically, the average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is, for example, 5 [μm] or more and 12 [μm] or less. In this case, the average particle diameter of the positive electrode active material

particles Pd contained in the deep layer portion **31***d* is, for example, 3 [μm] or more and 8 [μm] or less.
(32) For example, the average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* can be set to 6.4 [μm], and the average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d* can be set to 5.7 [μm]. The average particle diameters of the positive electrode active material particles Ps, Pd can be set to median diameters D50 of the positive electrode active material particles Ps, Pd measured by a particle size distribution measurement device using a laser diffraction scattering method and the like.
(33) A tap density of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is higher than, for example, a tap density of the positive electrode active material particles Pd contained in the deep layer portion **31***d*. More specifically, the tap density of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is, for example, 2.0 [g/cm.sup.3] or more and 2.8 [g/cm.sup.3] or less. Meanwhile, the tap density of the positive electrode active material particles Pd contained in the deep layer portion **31***d* is, for example, 1.5 [g/cm.sup.3] or more and 2.2 [g/cm.sup.3] or less.
(34) For example, the tap density of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* can be set to 2.2 [g/cm.sup.3], and the tap density of the positive electrode active material particles Pd contained in the deep layer portion **31***d* can be set to 2.0 [g/cm.sup.3]. The tap densities of the positive electrode active material particles Ps, Pd can be measured by a measuring method according to Japanese Industrial Standards JIS Z 2512:2012 or JIS R 1628:1997.
(35) As described above, the space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* is lower than the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*. More specifically, the space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* is, for example, 25 [%] or more and 14 [%] or less. In this case, the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d* is, for example, 28 [%] or more and 15 [%] or less. The space ratio between the positive electrode active material particles Ps and the space ratio between the positive electrode active material particles Pd can be controlled by, for example, adjusting the combined amount of the positive electrode active material particles Ps, Pd, an addition rate of the solvent, or the like in the slurry as the material of the superficial layer portion **31***s* and the deep layer portion **31***d*.
(36) Here, the space ratios of the positive electrode active material particles Ps, Ps are ratios of space between the positive electrode active material particles Ps, Ps in the positive electrode active material powder as an aggregation of the positive electrode active material particles Ps, Ps. The space ratios of the positive electrode active material particles Ps, Ps can be measured by, for example, an optical method in which a microscopy of a cross-sectional surface of a sample of the positive electrode mixture layer **31***b* is performed. More specifically, the space ratios of the positive electrode active material particles Ps, Ps can be obtained from, for example, areas of the positive electrode active material particles Ps, Ps per unit area in the cross-sectional surface of the sample of the superficial layer portion **31***s* and the deep layer portion **31***d*.
(37) Note that the positive electrode active material included in the positive electrode mixture layer **31***b* is not limited to the above-described layered nickel cobalt lithium manganate. As the positive electrode active material, for example, another lithium manganate having a spinel crystal structure, and a lithium manganese composite oxide partially replaced or doped with a metallic element can be used. Additionally, as the positive electrode active material, a lithium cobaltate or a lithium titanate that has a layered crystal structure, and a lithium-metal composite oxide in which a part of these substances is replaced or doped with a metallic element may be used.
(38) The binder used for the negative electrode mixture and the positive electrode mixture is not

limited to SBR or PVDF. As the binder, for example, a polymer, such as polytetrafluoroethylene (PTFE), polyethylene, polystyrene, polybutadiene, butyl rubber, nitrile rubber, polysulfide rubber, nitrocellulose, cyanoethyl cellulose, various kinds of latexes, acrylonitrile, vinyl fluoride, vinylidene fluoride, propylene fluoride, chloroprene fluoride, and acrylic-based resin, and a mixture of these substances can be used.
(39) The separators **33**, **34** are made of, for example, porous polyethylene resin or polypropylene resin, or a composite resin thereof and are interposed between the positive electrode **31** and the negative electrode **32** to electrically insulate them. In addition, the separators **33**, **34** are wound around an outside of the negative electrode **32** wound around the outermost periphery. Although the illustration is omitted, the wound body **30** may include a winding core for laminating the negative electrode **32**, the first separator **33**, the positive electrode **31**, and the second separator **34** to wind them.
(40) As the winding core, for example, one obtained by winding a resin sheet having a flexural rigidity higher than those of the positive electrode current collector foil **31***a*, the negative electrode current collector foil **32***a*, and the separators **33**, **34** can be used. The wound body **30** is configured such that dimensions of the negative electrode mixture layers **32***b* in the winding axis direction D (X-direction) are larger than dimensions of the positive electrode mixture layers **31***b* while the positive electrode mixture layers **31***b* are always sandwiched between the negative electrode mixture layers **32***b*.
(41) In the wound body **30**, as illustrated in FIG. **3**, each of the positive electrode current collector portion **31***c* of the positive electrode **31** and the negative electrode current collector portion **32***c* of the negative electrode **32** is wound to be laminated on one end and the other end in the winding axis direction D (X-direction). Furthermore, as illustrated in FIG. **2**, each of the positive electrode current collector portion **31***c* and the negative electrode current collector portion **32***c* is flatly bundled and is joined to the joint portions **42***a* of the extending portions **42** of the current collector plates **40** by, for example, ultrasonic joining or resistance welding.
(42) In the winding axis direction D (X-direction), dimensions of the separators **33**, **34** are larger than the dimensions of the negative electrode mixture layers **32***b*. However, the separators **33**, **34** have end portions each disposed at an inner position in the winding axis direction D (X-direction) with respect to end portions of the positive electrode current collector portion **31***c* and the negative electrode current collector portion **32***c*. Therefore, this does not obstruct bundling the positive electrode current collector portion **31***c* and the negative electrode current collector portion **32***c* and joining them to the joint portions **42***a* of the extending portions **42** of a positive electrode current collector plate **40**P and a negative electrode current collector plate **40**N, respectively.
(43) The base portions **41** of the current collector plates **40** are secured to the battery lid **12** via plate-shaped insulating members **14** and electrically connected to the external terminals **20**. More specifically, the connecting portions **22** of the external terminals **20** are inserted through, for example, through holes **13***a* of the gaskets **13**, the through holes **12***a* of the battery lid **12**, through holes **14***a* of the insulating members **14**, and through holes **41***a* of the base portions **41** of the current collector plates **40**. The connecting portions **22** of the external terminals **20** are plastically deformed to be caulked while the distal ends are radially expanded on lower surfaces of the base portions **41** of the current collector plates **40**.
(44) Thus, the external terminals **20** and the current collector plates **40** are secured in a state of being electrically connected to one another and being electrically insulated from the battery lid **12** via the gaskets **13** and the insulating members **14**, thus being assembled as a lid assembly. A material of the gasket **13** and the insulating member **14** is, for example, a resin having an electrical insulating property, such as polybutylene terephthalate, polyphenylene sulfide, and perfluoroalkoxy fluororesin.
(45) In the assembled lid assembly, the joint portions **42***a* of the extending portions **42** of the current collector plates **40** are joined to the respective laminated portions **35** of the positive

electrode current collector portion **31***c* and the negative electrode current collector portion **32***c* of the wound body **30**. Thus, the positive electrode **31** and the negative electrode **32**, which constitute the wound body **30**, are electrically connected to the external terminals **20** via the current collector plates **40**. The wound body **30** is joined to the current collector plates **40**, thereby being secured to the battery lid **12** via the current collector plates **40** and electrically connected to the external terminals **20**. In this state, the wound body **30** is covered with the resin insulating sheet **50** having an electrical insulating property, and inserted into the battery can **11** from the opening **11***a* of the battery can **11**.

(46) The insulating sheet **50** is formed of one sheet or a plurality of film members made of a material, for example, a synthetic resin, such as polypropylene. The insulating sheet **50** has dimensions and a shape that can cover approximately the whole wound body **30**, to which the current collector plates **40** have been joined, together with the current collector plates **40**. The insulating sheet **50** is interposed between the wound body **30** with the current collector plates **40** and the battery can **11**, so as to electrically insulate therebetween. The wound body **30** and the current collector plate **40** may be covered with a bag-shaped insulating film instead of the insulating sheet **50**.

(47) The wound body **30** is inserted into the battery can **11** from one curving portion **30***b* so as to have the winding axis direction D along the width direction (X direction) of the secondary battery **100**, and disposed to have the other curving portion **30***b* to be opposed to the battery lid **12**. Subsequently, as described above, the battery lid **12** is joined to the opening **11***a* of the battery can **11** over the whole circumference to constitute the battery container **10**. Subsequently, the electrolyte is injected into the battery container **10** via the liquid injection hole **16**, and the liquid injection hole **16** is sealed by joining the liquid injection plug **17**. At this time, appropriately adjusting a pressure inside the battery container **10** and a pressure outside the battery container **10** accelerates the replacement of the air with the electrolyte in the wound body **30**, thus allowing efficiently injecting the electrolyte into the battery container **10**.

(48) With the above-described configuration, in the secondary battery **100**, the external terminal **20** is connected to an external device via a not illustrated busbar joined to the external terminal **20**. The secondary battery **100** is charged by supplying an electric power to the positive electrode **31** and the negative electrode **32** of the wound body **30** via the external terminal **20** and the current collector plate **40**. The charged secondary battery **100** can supply the electric power to the external device from the positive electrode **31** and the negative electrode **32** of the wound body **30** via the current collector plate **40** and the external terminal **20**.

(49) The following describes actions of the secondary battery **100** of this embodiment based on the comparison with a conventional lithium ion secondary battery that includes a conventional positive electrode plate.

(50) In the conventional positive electrode plate, as described above, the smaller average particle diameter of the contained positive electrode active material particles the positive electrode active material layer has, the upper layer the positive electrode active material layer is disposed in. In the lithium ion secondary battery using this positive electrode plate, the positive electrode active material layer that is disposed in the upper layer of the positive electrode plate and has the small average particle diameter of the contained positive electrode active material particles is opposed to the negative electrode plate via the separator. The positive electrode active material layer in the upper layer of the positive electrode plate opposing the negative electrode plate via the separator increases in intercalation and deintercalation of lithium ions due to charge/discharge reaction of the battery compared with the positive electrode active material layer in the lower layer.

(51) However, for the positive electrode active material particles, cracking of the positive electrode active material particles easily occurs due to the intercalation and the deintercalation of the lithium ions as the average particle diameter relatively decreases. Therefore, in this conventional lithium ion battery, in the positive electrode active material layer of the upper layer, cracking occurs in the

positive electrode active material particles, thus possibly increasing the internal resistance value of the battery or decreasing the battery capacity.
(52) An on-vehicle secondary battery used for an electric storage device of a vehicle, such as an EV and an HEV, is desired to have further high output power. Furthermore, the on-vehicle secondary battery is desired to have an extended discharge power duration so as to allow extending the travel distance of the vehicle compared with the conventional one.
(53) In contrast, as described above, the secondary battery **100** of this embodiment is a lithium ion secondary battery, and includes the positive electrode **31**, the negative electrode **32**, and the separators **33**, **34**. The positive electrode **31** includes the positive electrode current collector foil **31***a* and the positive electrode mixture layers **31***b* disposed on the surfaces of the positive electrode current collector foil **31***a*. The positive electrode mixture layers **31***b* include the superficial layer portions **31***s* opposing the negative electrode **32** via the separators **33**, **34**, and the deep layer portions **31***d* disposed between the superficial layer portions **31***s* and the positive electrode current collector foil **31***a*. The average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is greater than the average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d*. The space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* is lower than the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*.
(54) Thus, since the average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is greater than the average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d* in the positive electrode mixture layer **31***b*, the cracking of the positive electrode active material particles Ps in the superficial layer portion **31***s* is suppressed. That is, the relatively large average particle diameter of the positive electrode active material particles Ps allows suppressing the cracking caused by the intercalation and the deintercalation of the lithium ions due to charge/discharge reaction in the secondary battery **100**.
(55) That is, the cracking of the positive electrode active material particles Ps in the positive electrode mixture layers **31***b*, which are opposed to the negative electrode **32** via the separators **33**, **34** and increased in intercalation and deintercalation of the lithium ions compared with the positive electrode active material particles Pd in the deep layer portion **31***d*, can be suppressed. Accordingly, the decrease in capacity of the secondary battery **100** can be suppressed, thereby allowing suppressing the increase in internal resistance of the secondary battery **100** to achieve the higher output power of the secondary battery **100**.
(56) Furthermore, by a shield effect in which the positive electrode active material particles Ps in the superficial layer portion **31***s* having the relatively high durability protect the positive electrode active material particles Pd in the deep layer portion **31***d* having the relatively low durability, the cracking of the positive electrode active material particles Pd in the deep layer portion **31***d* can be avoided. Accordingly, the secondary battery **100** of this embodiment can provide the lithium ion secondary battery that is excellent in durability and increased in capacity compared with the conventional lithium ion secondary battery.
(57) The average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d* is smaller than that of the positive electrode active material particles Ps contained in the superficial layer portion **31***s*. Therefore, by the positive electrode active material particles Pd that are contained in the deep layer portion **31***d* and have the relatively small average particle diameter and high capacity density, the decrease in capacity and the decrease in output power of the secondary battery **100** are avoided, thus allowing achieving an increased capacity and a higher output power of the secondary battery **100**. That is, the secondary battery **100** of this embodiment can provide the lithium ion secondary battery in which the capacity and the output power are increased by the relatively small positive electrode active material particles Pd

contained in the deep layer portion **31***d* while the durability of the secondary battery **100** is improved by the relatively large positive electrode active material particles Ps contained in the superficial layer portion **31***s*.
(58) Furthermore, since the space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* is lower than the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*, the positive electrode active material particles Ps are densely filled in the superficial layer portion **31***s*, thus avoiding missing the positive electrode active material particles Ps. Therefore, a local deterioration of the positive electrode mixture layer **31***b* caused by opening of a large hole in the superficial layer portion **31***s* is avoided, and the shield effect of the positive electrode active material particles Ps in the superficial layer portion **31***s* is sustained for a longer period, thus allowing improving the durability of the secondary battery **100**.
(59) Accordingly, the use of the secondary battery **100** of this embodiment for the electric storage device of the vehicle allows extending the discharge power duration in addition to the higher output power of the vehicle, such as an EV and an HEV, thereby allowing extending the travel distance of the vehicle compared with the conventional one.
(60) In the secondary battery **100** of this embodiment, as described above, for example, the tap density of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is higher than the tap density of the positive electrode active material particles Pd contained in the deep layer portion **31***d*.
(61) With this configuration, as described above, the space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* can be lowered compared with the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*. That is, when the tap density of the positive electrode active material particles Ps is higher than the tap density of the positive electrode active material particles Pd, the positive electrode active material particles Ps in the superficial layer portion **31***s* can be disposed with the density higher than that of the positive electrode active material particles Pd in the deep layer portion **31***d*, thereby avoiding missing the positive electrode active material particles Ps in the superficial layer portion **31***s*.
(62) In the secondary battery **100** of this embodiment, for example, the cobalt content of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is higher than the cobalt content of the positive electrode active material particles Pd contained in the deep layer portion **31***d*.
(63) With this configuration, in the positive electrode active material particles Ps contained in the superficial layer portion **31***s*, the layered structure can be more stably kept. Stably keeping the layered structure of the positive electrode active material particles Ps allows suppressing, for example, a cation mixing in which Ni is mixed in Li site. Therefore, excellent charge/discharge cycle characteristics can be obtained, thus allowing improving the durability of the positive electrode active material particles Ps contained in the superficial layer portion **31***s*.
(64) In the secondary battery **100** of this embodiment, as described above, for example, the tap density of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is 2.0 [g/cm.sup.3] or more and 2.8 [g/cm.sup.3] or less. In this case, the tap density of the positive electrode active material particles Pd contained in the deep layer portion **31***d* is, for example, 1.5 [g/cm.sup.3] or more and 2.2 [g/cm.sup.3] or less.
(65) With this configuration, fluidities of the positive electrode active material particles Ps, Pd can be ensured, thus allowing facilitating the handling of the positive electrode active material particles Ps, Pd. The space ratio between the positive electrode active material particles Ps in the superficial layer portion **31***s* can be more effectively lowered compared with the space ratio between the positive electrode active material particles Pd in the deep layer portion **31***d*. Accordingly, missing of the positive electrode active material particles Ps in the superficial layer portion **31***s* can be more

effectively avoided.
(66) In the secondary battery **100** of this embodiment, for example, as described above, the average particle diameter of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is 5 [μm] or more and 12 [μm] or less. In this case, the average particle diameter of the positive electrode active material particles Pd contained in the deep layer portion **31***d* is, for example, 3 [μm] or more and 8 [μm] or less.
(67) With this configuration, as described above, the cracking of the positive electrode active material particles Ps contained in the superficial layer portion **31***s* is suppressed, and the cracking of the positive electrode active material particles Pd in the deep layer portion **31***d* is avoided by the above-described shield effect, thus allowing sufficiently improving the durability of the secondary battery **100**. In addition, the capacity density of the positive electrode active material particles Pd contained in the deep layer portion **31***d* is improved, thus allowing more effectively achieving the increased capacity and the higher output power of the secondary battery **100**. Accordingly, for example, high requirements for the on-vehicle secondary battery can be satisfied.
(68) In the secondary battery **100** of this embodiment, for example as described above, the space ratio of the positive electrode active material particles Ps in the superficial layer portion **31***s* is 25 [%] or less and 14 [%] or more. In this case, the space ratio of the positive electrode active material particles Pd in the deep layer portion **31***d* is, for example, 28 [%] or less and 15 [%] or more.
(69) With this configuration, the positive electrode active material particles Ps can be densely filled in the superficial layer portion **31***s* of the positive electrode mixture layer **31***b*, thus allowing avoiding the missing of the positive electrode active material particles Ps in the superficial layer portion **31***s* with more certainty. Therefore, it is avoided that the shield effect is locally lost due to the missing of the positive electrode active material particles Ps in the superficial layer portion **31***s* having the larger average particle diameter than the positive electrode active material particles Pd in the deep layer portion **31***d*, thus allowing effectively avoiding the local deterioration of the positive electrode mixture layer **31***b*. Accordingly, the positive electrode active material particles Pd in the deep layer portion **31***d* having the small average particle diameter compared with the positive electrode active material particles Ps in the superficial layer portion **31***s* can be protected by the shield effect of the positive electrode active material particles Ps in the superficial layer portion **31***s*.
(70) As described above, this embodiment can provide the secondary battery **100** as the lithium ion secondary battery that is excellent in durability and increased in capacity compared with the conventional one.
(71) Next, with reference to FIG. **6** and FIG. **7**, a description will be given of a manufacturing method M of the lithium ion secondary battery according to one embodiment of this disclosure. FIG. **6** is a flowchart illustrating steps of the manufacturing method M of the lithium ion secondary battery of this embodiment. FIG. **7** is a schematic diagram illustrating a part of a manufacturing apparatus A used for the manufacturing method M of the lithium ion secondary battery of this embodiment.
(72) The manufacturing method M of the lithium ion secondary battery of this embodiment is a method, for example, for manufacturing the secondary battery **100** described above. More specifically, the manufacturing method M of the lithium ion secondary battery of this embodiment is a method for manufacturing the lithium ion secondary battery in which the positive electrode **31** is opposed to the negative electrode **32** via the separators **33**, **34**. The manufacturing method M of the lithium ion secondary battery of this embodiment is characterized by a manufacturing process of the positive electrode **31**. Therefore, the following describes the manufacturing process of the positive electrode **31** in detail, and the description of other processes will be appropriately omitted.
(73) As illustrated in FIG. **6**, the manufacturing method M of the lithium ion secondary battery of this embodiment includes an applying step S**1**, a drying step S**2**, a pressing step S**3**, a cutting step S**4**, and an electrode group manufacturing step S**5**. In addition, the manufacturing method M of the

lithium ion secondary battery of this embodiment includes, for example, an assembling step S**6**.
(74) The manufacturing apparatus A illustrated in FIG. **7** is an apparatus used in, for example, the applying step S**1** and the drying step S**2**. The manufacturing apparatus A includes, for example, a first slurry tank A**1**, a second slurry tank A**2**, pumps A**3**, A**4**, a first slit die A**5**, and a second slit die A**6**. The manufacturing apparatus A includes, for example, a supply roller A**7**, a collection roller A**8**, conveyance rollers A**9**, A**10**, and A**11**, a coating roller A**12**, and a dryer A**13**.
(75) The first slurry tank A**1** stores a first slurry SL**1** containing the first positive electrode active material particles Pd. The second slurry tank A**2** stores a second slurry SL**2** containing the second positive electrode active material particles Ps. The first slurry SL**1** is the material of the deep layer portion **31***d*, and the second slurry SL**2** is the material of the superficial layer portion **31***s*. As the first slurry SL**1** and the second slurry SL**2**, as described above, ones obtained by adding the conductive material, the binder, and the dispersing solvent to each of the positive electrode active material particles Pd and the positive electrode active material particles Ps and mixing them can be used.
(76) The pump A**3** is disposed to a channel of the first slurry SL**1** between the first slurry tank A**1** and the first slit die A**5**, and pressure-feeds the first slurry SL**1** from the first slurry tank A**1** to the first slit die A**5**. The pump A**4** is disposed to a channel of the second slurry SL**2** between the second slurry tank A**2** and the second slit die A**6**, and pressure-feeds the second slurry SL**2** from the second slurry tank A**2** to the second slit die A**6**.
(77) The first slit die A**5** discharges the first slurry SL**1** pressure-fed by the pump A**3** from a slit, thus applying it over the surface of the positive electrode current collector foil **31***a*. The second slit die A**6** is disposed rearward in a conveying direction of the positive electrode current collector foil **31***a* with respect to the first slit die A**5**, and applies the second slurry SL**2** pressure-fed by the pump A**4** over the surface of the first slurry SL**1** applied over the surface of the positive electrode current collector foil **31***a*.
(78) The supply roller A**7** rotates while supporting the positive electrode current collector foil **31***a* wound in a roll shape on the outer periphery, thereby supplying the positive electrode current collector foil **31***a*. The collection roller A**8** rotates, thereby winding the positive electrode current collector foil **31***a* with the positive electrode mixture layer **31***b* formed on the surface around the outer periphery and collecting it. The conveyance rollers A**9**, A**10**, and A**11** convey the positive electrode current collector foil **31***a* supplied from the supply roller A**7** to the coating roller A**12**, the dryer A**13**, and the collection roller A**8**.
(79) The coating roller A**12** rotates while supporting the positive electrode current collector foil **31***a* on the outer peripheral surface when the first slurry SL**1** and the second slurry SL**2** are applied over the positive electrode current collector foil **31***a*, thus conveying the positive electrode current collector foil **31***a* to the dryer A**13**. The dryer A**13** heats the first slurry SL**1** and the second slurry SL**2** applied over the surface of the positive electrode current collector foil **31***a*, and dries them.
(80) The applying step S**1** includes, for example, a step of applying a slurry as a material of the positive electrode mixture layer **31***b* over the positive electrode current collector foil **31***a* constituting the positive electrode **31**, and a step of applying a slurry as a material of the negative electrode mixture layer **32***b* over the negative electrode current collector foil **32***a* constituting the negative electrode **32**. In this embodiment, the applying step S**1** is characterized by the step of applying a slurry as a material of the positive electrode mixture layer **31***b* over the positive electrode current collector foil **31***a*. Therefore, the following describes the manufacturing process of the positive electrode **31**, and the description of the manufacturing process of the negative electrode **32** will be omitted.
(81) In this embodiment, the applying step S**1** includes a step of applying the first slurry SL**1** containing the first positive electrode active material particles Pd over the surface of the positive electrode mixture layer **31***b* and applying the second slurry SL**2** containing the second positive electrode active material particles Ps over the surface of the first slurry SL**1**. Here, the second

positive electrode active material particles Ps contained in the second slurry SL**2** as the material of the superficial layer portion **31***s* have the larger average particle diameter than the first positive electrode active material particles Pd contained in the first slurry SL**1** as the material of the deep layer portion **31***d*. The second positive electrode active material particles Ps have the high tap density compared with the first positive electrode active material particles Pd.
(82) More specifically, in the applying step S**1**, for example, the first slurry SL**1** and the second slurry SL**2** are applied over the positive electrode current collector foil **31***a* that is supplied from the supply roller A**7**, conveyed by the conveyance rollers A**9**, A**10**, and A**11**, and supported by the coating roller A**12** in the manufacturing apparatus A. In more detail, the first slurry SL**1** is pressure-fed by the pump A**3** from the first slurry tank A**1** to the first slit die A**5**. Subsequently, the first slurry SL**1** is discharged from the slit of the first slit die A**5**, thereby applying the first slurry SL**1** over the surface of the positive electrode current collector foil **31***a*.
(83) Furthermore, the second slurry SL**2** is pressure-fed by the pump A**4** from the second slurry tank A**2** to the second slit die A**6**. Subsequently, the second slurry SL**2** is discharged from the slit for the second slurry SL**2**, thereby applying the second slurry SL**2** over the surface of the first slurry SL**1** applied over the surface of the positive electrode current collector foil **31***a*. That is, in the applying step S**1**, the second slurry SL**2** is applied over the first slurry SL**1** before drying.
(84) In the applying step S**1**, for example, the first slurry SL**1** and the second slurry SL**2** are prepared such that the second slurry SL**2** has a viscosity lower than a viscosity of the first slurry SL**1**. Accordingly, the second slurry SL**2** applied over the first slurry SL**1** quickly and smoothly spreads to cover the surface of the first slurry SL**1**, thereby uniformly coating the surface of the first slurry SL**1** with the second slurry SL**2**.
(85) Note that, in the applying step S**1**, for example, the first slurry SL**1** and the second slurry SL**2** are applied excluding both ends in the width direction of the strip-shaped positive electrode current collector foil **31***a*. In addition, while the first slurry SL**1** and the second slurry SL**2** are applied using the first slit die A**5** and the second slit die A**6** in the manufacturing apparatus A illustrated in FIG. **7**, the manufacturing apparatus A is not limited to this configuration. For example, the first slurry SL**1** and the second slurry SL**2** laminated in advance may be simultaneously discharged from one slit die. After completing the applying step S**1**, the drying step S**2** starts.
(86) The drying step S**2** is a step of drying the first slurry SL**1** applied over the surface of the positive electrode current collector foil **31***a* and the second slurry SL**2** applied over the surface of the first slurry SL**1**. In this step, the deep layer portion **31***d* containing the first positive electrode active material particles Pd is formed on the surface of the positive electrode current collector foil **31***a*, and the superficial layer portion **31***s* containing the second positive electrode active material particles Ps is formed on the surface of the deep layer portion **31***d*.
(87) More specifically, in the drying step S**2**, for example, the positive electrode current collector foil **31***a* is conveyed by the conveyance rollers A**9**, A**10**, and A**11** of the manufacturing apparatus A, thus sending the positive electrode current collector foil **31***a* with the first slurry SL**1** and the second slurry SL**2** applied over its surface into the dryer A**13**. For example, the dryer A**13** heats the first slurry SL**1** and the second slurry SL**2** applied over the surface of the positive electrode current collector foil **31***a* to vaporize the solvent by the hot wind, thus simultaneously drying them. A device to suction and collect the vaporized solvent may be added to enhance the efficiency of the drying step S**2**.
(88) At this time, fine gaps or voids are provided between the positive electrode active material particles Pd contained in the first slurry SL**1**, between the positive electrode active material particles Ps contained in the second slurry SL**2**, and between the particles of the conductive auxiliary agent. The gaps or the voids have the size in the close-packed arrangement determined by the average particle diameters of the positive electrode active material particles Pd and the positive electrode active material particles Ps. The gaps or the voids can be controlled by a mixing ratio of the positive electrode active material particles Pd and the positive electrode active material

particles Ps, the conductive auxiliary agent, the solvent, and the like in preparing the first slurry SL**1** and the second slurry SL**2**.
(89) In FIG. **7**, the positive electrode mixture layer **31***b* including the deep layer portion **31***d* and the superficial layer portion **31***s* is formed by applying the first slurry SL**1** and the second slurry SL**2** over only one surface of the positive electrode current collector foil **31***a*. Subsequently, for example, the positive electrode current collector foil **31***a* with the positive electrode mixture layer **31***b* formed on its one surface is supported to the outer periphery of the supply roller A**7**, and the first slurry SL**1** and the second slurry SL**2** are applied over the other surface of the positive electrode current collector foil **31***a* and dried, thus allowing forming the positive electrode mixture layers **31***b* on both surfaces of the positive electrode current collector foil **31***a*. After completing the drying step S**2**, the pressing step S**3** starts.
(90) The pressing step S**3** is a step of pressurizing a laminated body L in which the deep layer portion **31***d* is disposed on the surface of the positive electrode current collector foil **31***a* and the superficial layer portion **31***s* is disposed on the surface of the deep layer portion **31***d* through the drying step S**2**. In the pressing step S**3**, for example, the laminated body L is sandwiched between a pair of press rollers and pressed. After completing the pressing step S**3**, the cutting step S**4** starts.
(91) The cutting step S**4** is a step of cutting the laminated body L that has undergone the pressing step S**3** to obtain the positive electrode **31** that includes the positive electrode mixture layer **31***b* including the superficial layer portion **31***s* and the deep layer portion **31***d*. In the cutting step S**4**, for example, the laminated body L in which the positive electrode mixture layer **31***b* is disposed on the surface of the positive electrode current collector foil **31***a* excluding both ends in the width direction of the strip-shaped positive electrode current collector foil **31***a* is cut at the center in the width direction of the positive electrode current collector foil **31***a*. Accordingly, as illustrated in FIG. **3** and FIG. **4**, the positive electrode **31** that includes the positive electrode current collector portion **31***c* on one end in the width direction is obtained. After completing the cutting step S**4**, the electrode group manufacturing step S**5** starts.
(92) The electrode group manufacturing step S**5** is a step of causing the positive electrode **31** to oppose the negative electrode **32** via the separators **33**, **34**, and causing the superficial layer portion **31***s* of the positive electrode mixture layer **31***b* to oppose the negative electrode **32** via the separators **33**, **34**. More specifically, in the electrode group manufacturing step S**5**, the separator **33** and the separator **34** are wound around a rotation shaft of a winding machine, and the positive electrode **31** and the negative electrode **32** are each sandwiched between the separators **33**, **34** and wound.
(93) Accordingly, as illustrated in FIG. **3**, **30** in which the electrode wound around the innermost periphery and the outermost periphery is the negative electrode **32** and the first separator **33** is further wound around the outer periphery of the negative electrode **32** wound around the outermost periphery can be manufactured. Note that, in the electrode group manufacturing step S**5**, a winding core may be attached to the rotation shaft of the winding machine, starting end portions of the separator **33** and the separator **34** may be welded to the winding core, and the separator **33**, the negative electrode **32**, the separator **34**, and the positive electrode **31** may be laminated and wound. After completing the electrode group manufacturing step S**5**, for example, the assembling step S**6** starts.
(94) The assembling step S**6** is a step of assembling the battery container **10**, the external terminal **20**, the wound body **30**, the current collector plate **40**, and the insulating sheet **50** to constitute the secondary battery **100**. Specifically, as illustrated in FIG. **2**, the connecting portions **22** of the external terminals **20** are inserted through the through holes **13***a* of the gaskets **13**, the through holes **12***a* of the battery lid **12**, the through holes **14***a* of the insulating members **14**, and the through holes **41***a* of the current collector plates **40**, and distal ends of the connecting portions **22** of the external terminals **20** are plastically deformed to form caulking portions.
(95) Accordingly, the external terminal **20**, the gasket **13**, the insulating member **14**, and the current

collector plate **40** are integrally mounted to the battery lid **12**, and the external terminal **20** and the current collector plate **40** are electrically connected. The battery lid **12** is electrically insulated from the external terminals **20** and the current collector plates **40** by the gaskets **13** and the insulating members **14**. Furthermore, the laminated portion **35** of the wound body **30** is joined to the extending portions **42** of the current collector plates **40**.
(96) Accordingly, the positive electrode external terminal **20**P is connected to the positive electrode current collector portion **31***c* of the positive electrode **31** via the positive electrode current collector plate **40**P, and the negative electrode external terminal **20**N is connected to the negative electrode current collector portion **32***c* of the negative electrode **32** via the negative electrode current collector plate **40**N. The lid assembly in which the wound body **30** is supported to the battery lid **12** via the current collector plates **40** can be constituted. Subsequently, the insulating sheet **50** is wound around the wound body **30** and the current collector plates **40** to cover the wound body **30** and the current collector plates **40** with the insulating sheet **50**.
(97) The wound body **30** and the current collector plates **40** covered with the insulating sheet **50** are inserted through the opening **11***a* of the battery can **11**, thus housing them in the battery can **11**. Subsequently, in a state where the opening **11***a* of the battery can **11** is covered with the battery lid **12**, the whole circumference of the battery lid **12** is joined to the battery can **11** by a laser beam welding and the like, thus constituting the battery container **10**. Subsequently, a nonaqueous electrolyte is injected into the battery container **10** from the liquid injection hole **16** of the battery lid **12**, and the liquid injection plug **17** is joined to the liquid injection hole **16** to seal the battery container **10** by the laser beam welding and the like. Accordingly, the secondary battery **100** illustrated in FIG. **1** can be manufactured.
(98) As described above, the manufacturing method M of the lithium ion secondary battery of this embodiment is a method for manufacturing the lithium ion secondary battery in which the positive electrode **31** is opposed to the negative electrode **32** via the separators **33**, **34**. As described above, the manufacturing method M of the lithium ion secondary battery of this embodiment includes the following steps. The applying step S**1** of applying the first slurry SL**1** containing the first positive electrode active material particles Pd over the surface of the positive electrode current collector foil **31***a* and applying the second slurry SL**2** containing the second positive electrode active material particles Ps over the surface of the first slurry SL**1**, the second positive electrode active material particles Ps having the larger average particle diameter and the higher tap density than the first positive electrode active material particles Pd. The drying step S**2** of drying the first slurry SL**1** applied over the surface of the positive electrode current collector foil **31***a* and the second slurry SL**2** applied over the surface of the first slurry SL**1**, forming the deep layer portion **31***d* containing the first positive electrode active material particles Pd on the surface of the positive electrode current collector foil **31***a*, and forming the superficial layer portion **31***s* containing the second positive electrode active material particles Ps on the surface of the deep layer portion **31***d*. The pressing step S**3** of pressurizing the laminated body L in which the deep layer portion **31***d* is disposed on the surface of the positive electrode current collector foil **31***a* and the superficial layer portion **31***s* is disposed on the surface of the deep layer portion **31***d* through the drying step S**2**. The cutting step S**4** of cutting the laminated body L that has undergone the pressing step S**3** to obtain the positive electrode **31** that includes the positive electrode mixture layer **31***b* including the superficial layer portion **31***s* and the deep layer portion **31***d*. The electrode group manufacturing step S**5** of causing the positive electrode **31** to oppose the negative electrode **32** via the separators **33**, **34**, and causing the superficial layer portion **31***s* of the positive electrode mixture layer **31***b* to oppose the negative electrode **32** via the separators **33**, **34**.
(99) That is, in the manufacturing method M of the lithium ion secondary battery of this embodiment, as described above, the second slurry SL**2** is laminated on the first slurry SL**1** in the applying step S**1**, and the first slurry SL**1** and the second slurry SL**2** are simultaneously dried in the drying step S**2**. Subsequently, the laminated body L that has undergone the drying step S**2** is

pressurized in the pressing step S**3**. Therefore, the secondary battery **100** manufactured by the manufacturing method M of the lithium ion secondary battery of this embodiment has an unevenness on an interface between the superficial layer portion **31***s* and the deep layer portion **31***d* included in the positive electrode mixture layer **31***b* of the positive electrode **31**. Accordingly, the deep layer portion **31***d* and the superficial layer portion **31***s* are integrated, thus allowing manufacturing the secondary battery **100** that is excellent in durability and increased in capacity compared with the conventional one as described above.

(100) Meanwhile, it is considered that after applying the first slurry SL**1** over the surface of the positive electrode current collector foil **31***a*, the first slurry SL**1** is dried to form the deep layer portion **31***d*, the deep layer portion **31***d* is pressed to be pressurized, and subsequently, the second slurry SL**2** is applied over the surface of the deep layer portion **31***d*. In this case, the surface of the deep layer portion **31***d* is pressed to be flattened. Therefore, when the second slurry SL**2** applied over the surface of the deep layer portion **31***d* is dried to form the superficial layer portion **31***s*, and the superficial layer portion **31***s* is pressed to be pressurized, the interface between the superficial layer portion **31***s* and the deep layer portion **31***d* becomes a flat surface without an unevenness as described above. However, this method not only reduces the productivity, but also possibly causes the superficial layer portion **31***s* and the deep layer portion **31***d* to be insufficiently joined. Accordingly, the secondary battery **100** is preferably manufactured by the above-described manufacturing method M of the lithium ion secondary battery.

(101) While the embodiments of the lithium ion secondary battery and the manufacturing method thereof according to the disclosure have been described in detail using the drawings, the specific configuration is not limited to the embodiments. Design changes and the like within a scope not departing from the gist of the disclosure are included in the disclosure.

REFERENCE SIGNS LIST

(102) **31** Positive electrode **31***a* Positive electrode current collector foil **31***b* Positive electrode mixture layer **31***d* Deep layer portion **31***s* Superficial layer portion **32** Negative electrode **33** Separator **33** Separator L Laminated body M Manufacturing method of lithium ion secondary battery Pd Positive electrode active material particles (first positive electrode active material particles) Ps Positive electrode active material particles (second positive electrode active material particles) S**1** Applying step S**2** Drying step S**3** Pressing step S**4** Cutting step S**5** Electrode group manufacturing step SL**1** First slurry SL**2** Second slurry

Claims

1. A lithium ion secondary battery comprising: a positive electrode; a negative electrode; and a separator, wherein the positive electrode includes a positive electrode current collector foil and a positive electrode mixture layer disposed on a surface of the positive electrode current collector foil, wherein the positive electrode mixture layer includes a superficial layer portion and a deep layer portion, the superficial layer portion opposes the negative electrode via the separator, and the deep layer portion is disposed between the superficial layer portion and the positive electrode current collector foil, wherein the superficial layer portion contains positive electrode active material particles having an average particle diameter larger than an average particle diameter of positive electrode active material particles contained in the deep layer portion, wherein a space ratio between the positive electrode active material particles in the superficial layer portion is lower than a space ratio between the positive electrode active material particles in the deep layer portion; wherein a cobalt content of the positive electrode active material particles contained in the superficial layer portion is higher than a cobalt content of the positive electrode active material particles contained in the deep layer portion; wherein the positive electrode active material particles contained in at least one of the superficial layer portion and the deep layer portion are composed of a nickel cobalt lithium manganate and wherein a ratio of cobalt metal elements constituting anions

of the positive electrode active material particles composed of the nickel cobalt lithium manganate is more than 18% and less than 40% in terms of molar percentage with respect to all metals constituting anions.

2. The lithium ion secondary battery according to claim 1, wherein the average particle diameter of the positive electrode active material particles contained in the superficial layer portion is 5 μm or more and 12 μm or less, and wherein the average particle diameter of the positive electrode active material particles contained in the deep layer portion is 3 μm or more and 8 μm or less.

3. The lithium ion secondary battery according to claim 1, wherein the space ratio in the superficial layer portion is 25% or more and 14% or less, and wherein the space ratio in the deep layer portion is 28% or more and 15% or less.

4. A method for manufacturing a lithium ion secondary battery in which a positive electrode is opposed to a negative electrode via a separator, the method comprising: an applying step of applying a first slurry over a surface of a positive electrode current collector foil and applying a second slurry over a surface of the first slurry, the first slurry containing first positive electrode active material particles, the second slurry containing second positive electrode active material particles having a larger average particle diameter and a higher tap density than the first positive electrode active material particles; a drying step of drying the first slurry applied over the surface of the positive electrode current collector foil and the second slurry applied over the surface of the first slurry, forming a deep layer portion containing the first positive electrode active material particles on the surface of the positive electrode current collector foil, and forming a superficial layer portion containing the second positive electrode active material particles on a surface of the deep layer portion; a pressing step of pressurizing a laminated body in which the deep layer portion is disposed on the surface of the positive electrode current collector foil and the superficial layer portion is disposed on the surface of the deep layer portion through the drying step; a cutting step of cutting the laminated body that has undergone the pressing step to obtain a positive electrode that includes a positive electrode mixture layer including the superficial layer portion and the deep layer portion; an electrode group manufacturing step of causing the positive electrode to oppose the negative electrode via the separator, and causing the superficial layer portion of the positive electrode mixture layer to oppose the negative electrode via the separator; wherein a cobalt content of the positive electrode active material particles contained in the superficial layer portion is higher than a cobalt content of the positive electrode active material particles contained in the deep layer portion; wherein the positive electrode active material particles contained in at least one of the superficial layer portion and the deep layer portion are composed of a nickel cobalt lithium manganate and wherein a ratio of cobalt metal elements constituting anions of the positive electrode active material particles composed of the nickel cobalt lithium manganate is more than 18% and less than 40% in terms of molar percentage with respect to all metals constituting anions.

5. The lithium ion secondary battery of claim 1, wherein the positive electrode active material has the formula Li(Ni.sub.xCo.sub.yMn.sub.1-x-y)O.sub.2, wherein 0.30<x≤0.85 and 0.18<y<0.40.

6. A method for manufacturing a lithium ion secondary battery of claim 4, wherein the positive electrode active material has the formula Li(Ni.sub.xCo.sub.yMn.sub.1-x-y)O.sub.2, wherein 0.30<x≤0.85 and 0.18<y<0.40.